

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Mineral resources (the lithosphere) - Set 1

Mark scheme

AQA spec 3.2.3; The physical environment; textbook pp. 130-147

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Igneous intrusions (batholiths) of magma cool within the crust, heating groundwater to produce superheated, pressurised water • The hot water dissolves minerals from the surrounding rock and the mineral-rich solutions travel along fissures / cracks away from the batholith • As the solutions cool the dissolved minerals crystallise in order of solubility (least soluble first) - fractional crystallisation • So different metals are separated and concentrated in veins, eg tin, copper, lead, silver, gold, arsenic (SW England batholith)	4
01.2	• Dissolved iron compounds in ancient seas were oxidised by the oxygen released by early photosynthetic organisms (2.5-1.8 billion years ago) • Iron oxides are insoluble so they were deposited as sediments (haematite, magnetite)	2
01.3	• Flowing water carries solid particles; the ability to carry them depends on the water velocity and the density of the solids • As velocity falls, dense minerals (gold, tin ore) are deposited first while lighter material is carried further, concentrating the heavy minerals in particular places	2
01.4	<i>Any 2 from:</i> • Limestone / chalk from the shells of marine organisms • Coal from terrestrial vegetation • Crude oil and natural gas from marine organisms (plankton)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 132-134; AQA spec 3.2.3.2.	
02.1	• As the (linear) purity / grade of a mineral deposit decreases, the amount of the mineral present increases exponentially / logarithmically	1
02.2	• $2400 / 40 = 60$ times	1

Question	Marking guidance	Marks
02.3	• Reserves: the amount that can be exploited now, economically, with existing technology (deposits above the cut-off ore grade) • Resource: all the material that could be exploited technically and economically now or in the future (with realistic price rises or new technology), including deep, low-grade or protected deposits • The resource is finite but the reserves change: they rise if prices rise or technology improves, and fall if prices fall	3
02.4	• Lowering the cut-off ore grade includes deposits of lower purity in the reserves • Because the amount of material rises exponentially as purity falls, a small drop in COOG adds a very large mass (eg 0.5-1.0% has 4x the copper of 1.0-2.0%)	2
02.5	• The total quantity (stock / resource base) is far greater than could ever be used • But most is at purities too low or in locations too inaccessible for current economic exploitation • The challenge is developing technologies (low-energy extraction such as bioleaching, phytomining; better exploration) that make low-grade deposits exploitable; energy needs rise rapidly as ore grade falls	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 135-136; AQA spec 3.2.3.3.	
03.1	• Gravimetry: gravimeters detect variations in gravity caused by differences in rock density / mass (metal ores usually denser than sedimentary rock) • Magnetometry • Seismic survey: sound / shock waves from controlled explosions or a seismic vibrator; echoes recorded by geophones give the depth, density and shape of rock strata • Resistivity	4
03.2	• Trial drilling is the most expensive technique per site • Remote sensing (satellite, aircraft, ground surveys) covers large areas at low cost per unit area and identifies promising sites • Drilling is then targeted where deposits are likely; it is the only method that produces actual rock samples for chemical analysis to confirm composition and purity (proven reserves)	3
03.3	• Different minerals emit / reflect infrared at different wavelengths • Multispectral sensors detect these wavelengths over the whole surface (eg Landsat 8 every 16 days) giving information on surface mineral composition	2
03.4	• Results are available immediately in the field, saving time waiting for laboratory analysis	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 137; AQA spec 3.2.3.4.	
04.1	• The lowest ore purity that can be mined economically with existing technology • Deposit X (0.9% is above 0.4%; Y at 0.35% is part of the resource, not the reserves)	2

Question	Marking guidance	Marks
04.2	• X has higher grade so less rock mined, less spoil, less energy and pollution per tonne of copper • But X is deep: costs rise more than proportionally with depth (sides cannot be vertical so much more rock removed; more water inflow and pumping) • X has hard overburden requiring blasting (cost, noise, dust) and high rainfall increasing drainage costs • X is far from transport infrastructure so transport costs are high • Y is shallow with loose overburden (cheap to remove but landslide risk so gentler slopes / larger land take), near a railway, low rainfall • Y's grade is below the COOG so it is not viable now but would become viable if copper prices rose or extraction technology improved (eg bioleaching); both sulfides risk acid mine drainage	6
04.3	<i>Any 2 from:</i> • Higher market value / price of the metal • Improved extraction technology • Lower energy costs	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 138-139; AQA spec 3.2.3.5.	
05.1	• Minerals can only be exploited where they were deposited (no choice of location), unlike housing, roads or ports • Land take is larger than the mine void: buildings, access roads, overburden and spoil heaps, buffer zones; open-cast mines take far more land than deep mines	2
05.2	• Dust: water sprays on roads / blasting areas make particles heavier and clump so they settle • Noise: baffle mounds / embankments absorb and deflect; blasting at set, predictable daytime hours; quieter machinery • Turbid water: sedimentation lagoons where water stands until solids settle before discharge • Leachate: collect drainage water and pass through crushed limestone to neutralise acid and immobilise toxic metals • Amenity: landscaping and tree planting to screen the mine; restoration plans agreed as a condition of planning permission	5
05.3	• Landscaping to reduce gradients, compaction, drainage to prevent waterlogging, adding soil / nutrients and planting vegetation whose roots bind the surface • At Aberfan a waterlogged coal-mine spoil heap became unstable and slid into the village • Killing 144 people, mainly children in the primary school	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 139-142; AQA spec 3.2.3.6.	
06.1	• The void fills with water naturally; wetlands are uncommon / valuable habitats in most regions • The new habitat may have greater wildlife value than the farmland lost; low cost compared with filling and levelling	2
06.2	• Housing: ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer) • Farmland: landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced)	2

Question	Marking guidance	Marks
06.3	Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost; transplanted plants often fail	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6.	
07.1	- Acidophilic bacteria (eg Acidithiobacillus) oxidise sulfide ores, producing sulfuric acid which dissolves the metal ions - Leachate (rainwater / recirculated solution) percolating through the spoil heap is collected; recirculation increases the metal concentration - Copper is recovered from solution by electrolysis or by displacement with scrap iron (iron is more reactive so copper is deposited as solid metal) - Smelting needs chemical reduction at high temperatures; energy needed rises rapidly as ore grade falls, so low-grade ores are uneconomic - Bioleaching works at low temperatures with low energy inputs, so it can exploit low-grade deposits and wastes (increasing reserves) - mimics natural low-temperature processes	5
07.2	- Plants (hyperaccumulators) absorb metal ions from soil / water and concentrate them in their leaves - The vegetation is harvested and incinerated - The concentrated metal in the ash is dissolved in acid and separated by electrolysis; also decontaminates polluted sites	3
07.3	- Uranium ions dissolved in seawater adsorb onto the surface of synthetic or natural polymers (lignin from wood, chitin from shrimp shells) - The polymers are collected and the uranium recovered, avoiding mining / high-energy extraction	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 143-145; AQA spec 3.2.3.7.	
08.1	- Metal-rich lumps 5-10 cm across on the deep seabed (4000-5000 m), containing about 30% manganese plus nickel, copper, cobalt, iron, titanium - Exploitation would be very expensive (remotely operated collectors, pumping to the surface) / no viable technology yet - International agreement on ownership of the seabed and its resources is needed	3
08.2	- Collectors disturb the seabed and kill benthic organisms living on it - Separating nodules from sediment creates plumes that increase turbidity with unknown consequences - Re-deposition of sediment covers and kills filter feeders and other benthic organisms; slow recovery in the deep sea	3
08.3	- Used in small amounts in many essential technologies (smartphones, hybrid vehicles, wind turbines, fibre optics, LEDs) - Exploitable land deposits are limited and mainly in China; chemical separation is complex and produces large amounts of toxic waste - Bacterial adsorption: dissolved rare earth ions adsorb onto the cell surfaces of bacteria (<i>Bacillus subtilis</i>) and are washed off at specific pHs to separate them - Deep sea sediments contain relatively high concentrations but no viable extraction method yet	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 145-146; AQA spec 3.2.3.7.	